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Language Models as Explainable Classifiers for Skin Lesions

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Background Information

- Skin cancer is the most common form of cancer in the U.S., with over 20% of Americans developing it (American Academy of Dermatology, 2025)
- AI algorithms can accurately detect skin cancer (Tschandl et al., 2019)
  - They are not trusted due to lack of explainability (Von Eschenbach, 2021)
- Vision Language Models can explain their diagnoses.
- Reinforcement Learning can boost model accuracy on verifiable tasks (Shao et al., 2024).

Project Goals

Accuracy

The model should accurately classify skin moles as malignant or benign

Explainability

The model should generate valid explanations for its diagnoses

Experimental Design

- Trained using Reinforcement Learning with Verifiable Rewards
- Low-Rank Adapters (Hu et al., 2021) were used for efficiency
- Trained on images of skin moles tagged as benign or malignant from a Kaggle mirror of ISIC data
- Prompts were generated using a lattice approach to introduce diversity
- Correctness and formatting rewards were utilized

Results

Notes on Results

- Direct comparison with Model Dermatology (Han et al., 2020) was not performed because the system requires a different evaluation pipeline and was trained and benchmarked on different datasets from the model being presented.
- Instead of evaluating on separate baselines, performance was compared to the original, untrained model to measure the improvements caused by the training process.
- The y-axes of charts show decimal values. To get percentages, simply multiply by 100.

Results pt. 1 - Accuracy on Test Set

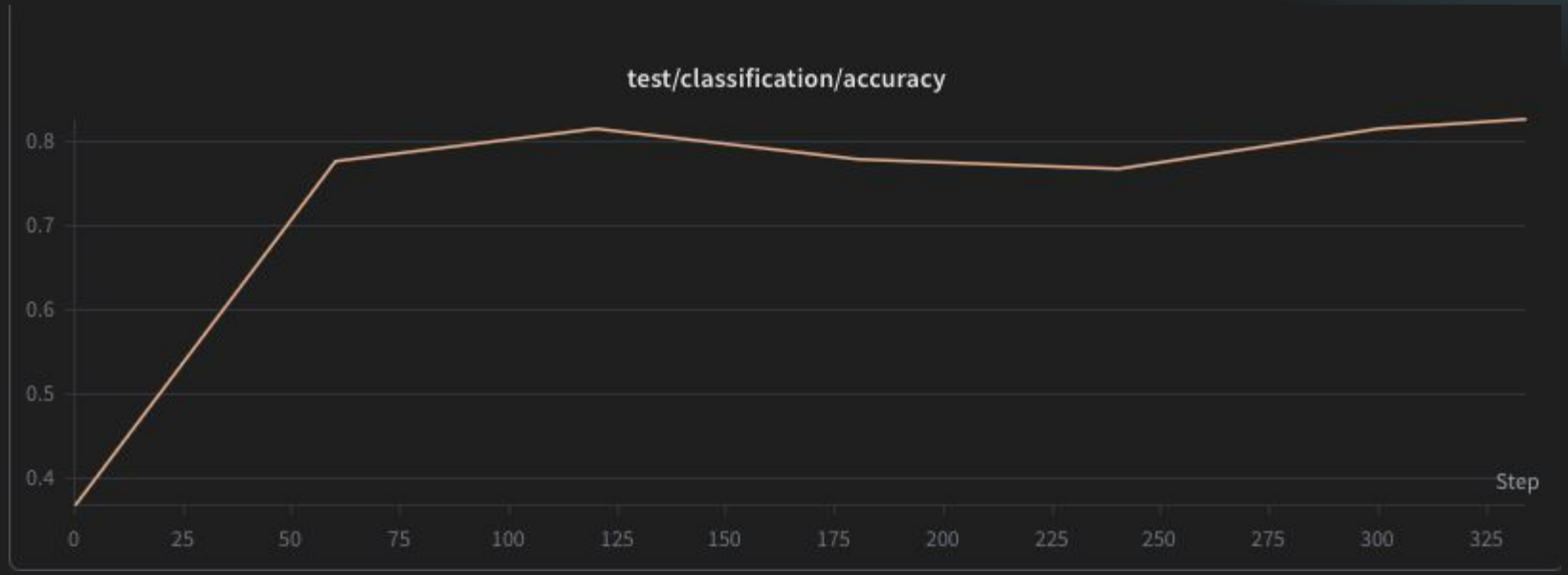


Figure 1. Over the course of training, accuracy rises significantly from 36.7% initially to 82.8% by the end of training.

Results pt. 2 – Precision on Test Set

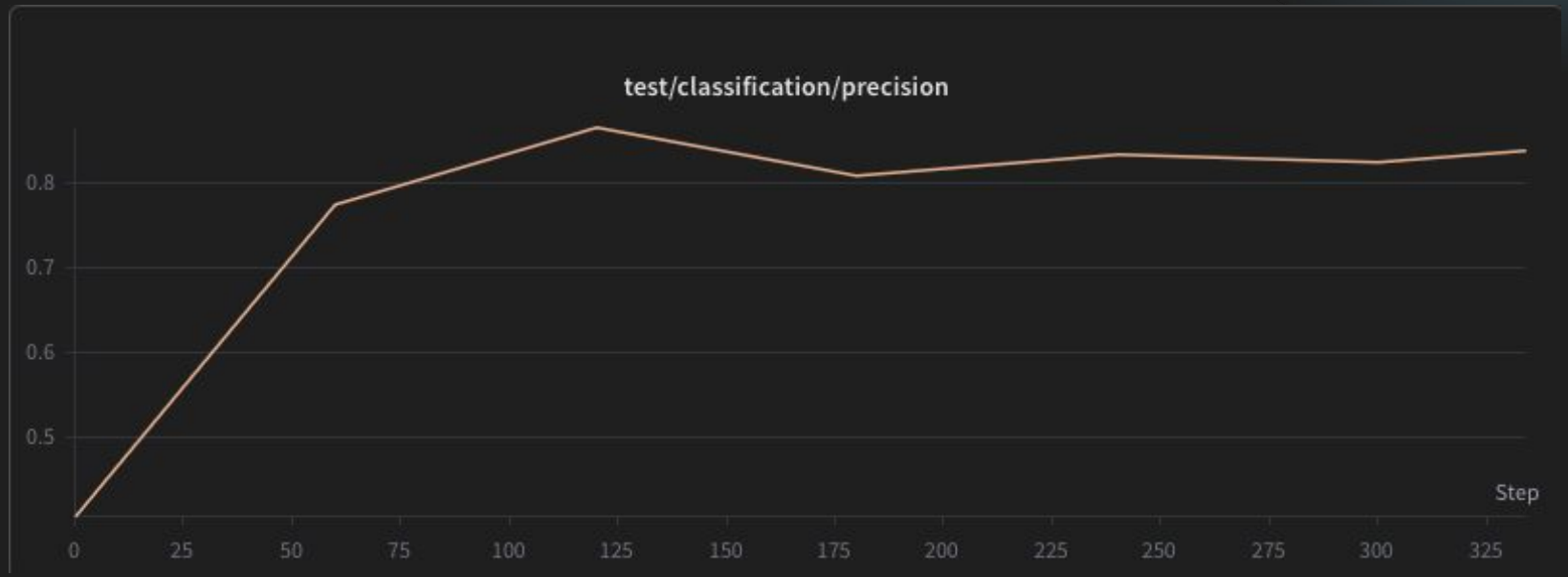


Figure 2. Precision measures what percent of cases classified by the model as positive were actually positive. It rises from 40.7% to 83.8%.

Results pt. 3 – Recall on Test Set

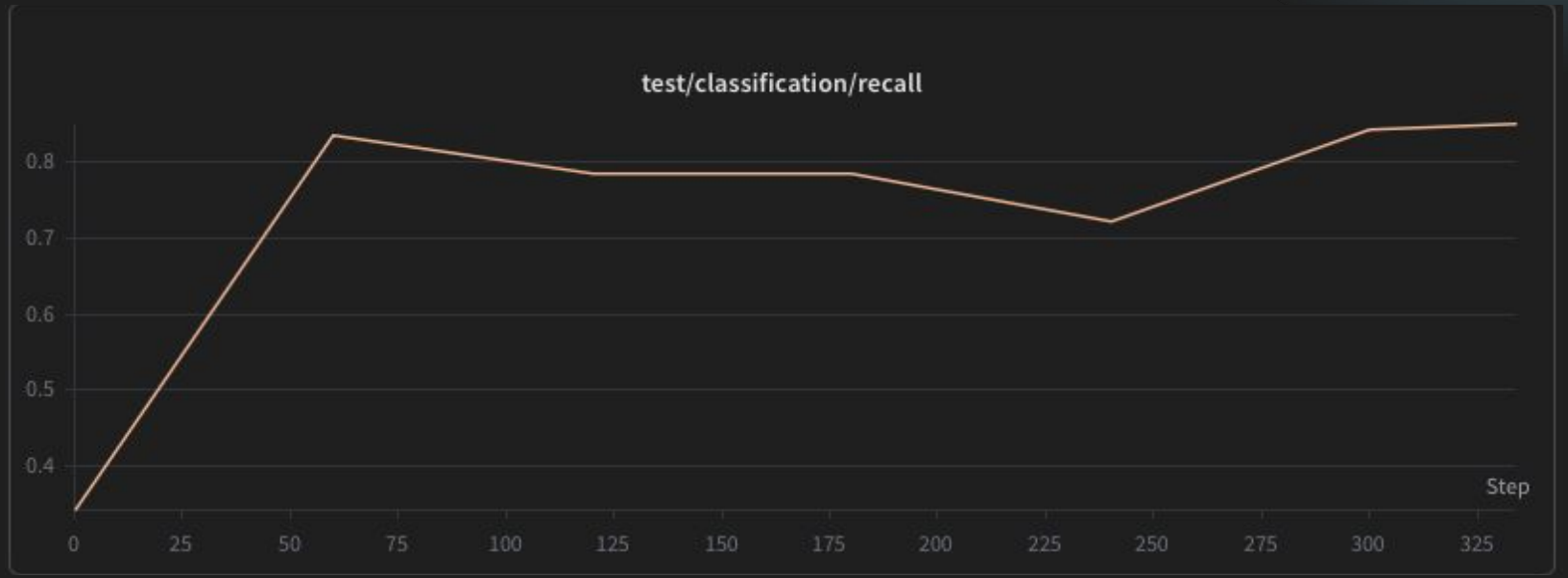


Figure 3. Recall measures what percent of positive cases were correctly identified by the model. It rises from 34.3% to 85.0%.

Results pt. 4 - F1 on Test Set

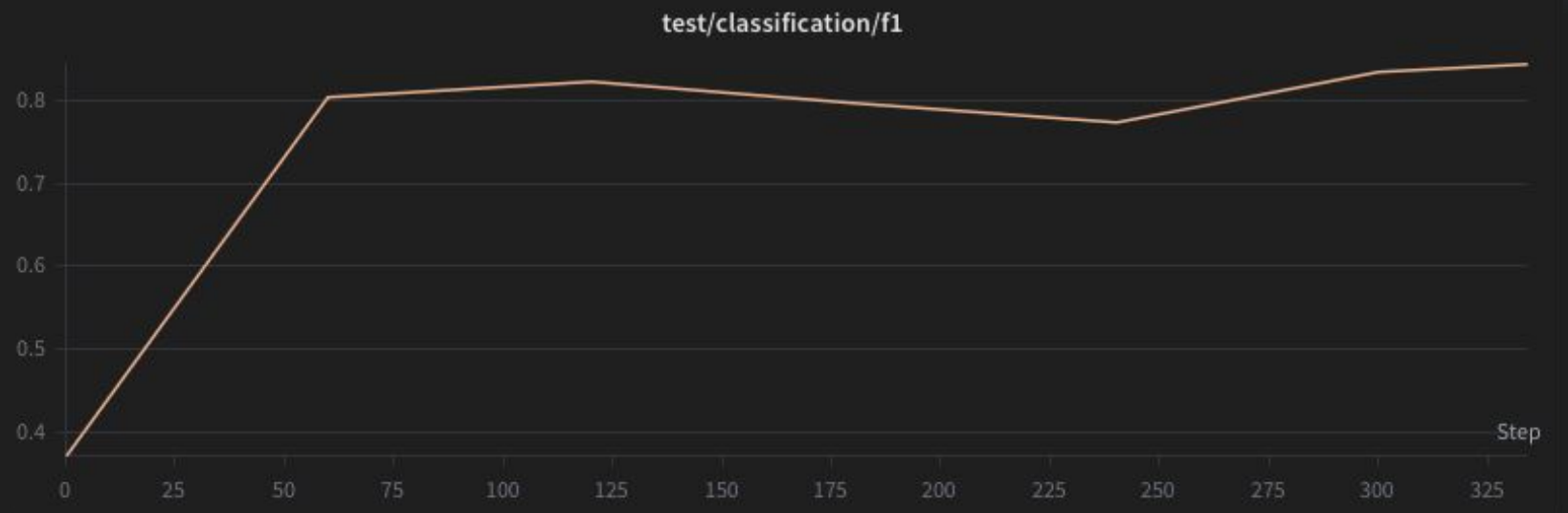


Figure 4. The harmonic mean of precision and recall, measuring how well a model balances catching real positives while avoiding false positives. It rises from 37.3% to 84.4%.

Results pt. 5 - McNemar test on Test Set

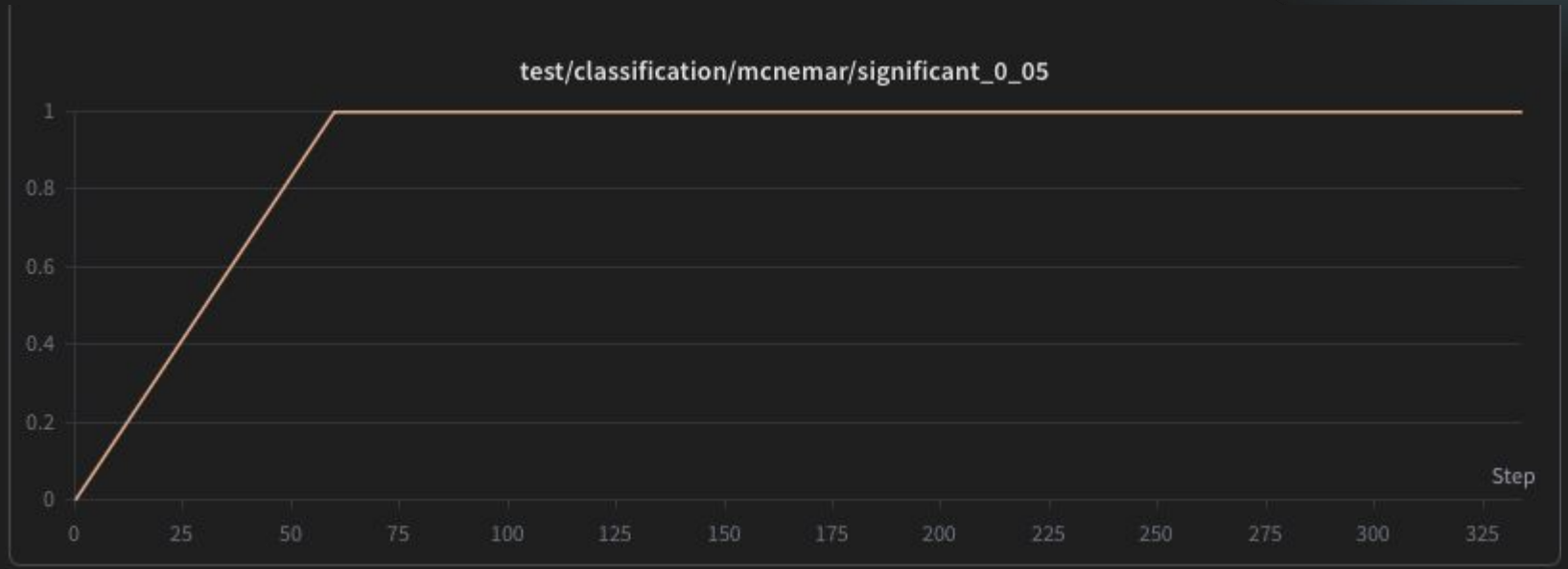


Figure 5. The McNemar test measures if training was statistically significant or due to random variation. Statistical significance ($p<0.05$) was achieved after only 17.9% of training, and at the end of training the p-value was 1.38e-28, indicating overwhelming evidence for statistical significance.

Results pt. 6 – Explanation Quality

Prompt: Patient presents with a lesion. What is your impression? Age: 55.0. Gender: male.

Model Response:

Model reasoning, summarized for brevity:

- Lesion appears as a round, well-circumscribed spot
- Lesion has a smooth border.
- The color is uniform throughout the structure, and the shape is symmetrical.
- These characteristics suggest a benign nevus.

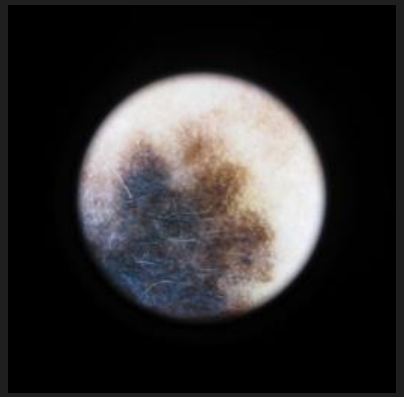
Final output:

Diagnosis: Benign nevus.

Visual evidence: The nevus is round, symmetrical, and has a consistent shade of brown. These features are typical of a harmless mole. Therefore, the patient should be advised on skin protection and regular monitoring for any changes in the lesion over time.

Figure 6. The model correctly identifies the image as benign, and references 3 of 5 ABCDE criteria: symmetry (A), border (B), and color (C). It is impossible for the model to reference the other 2 criteria (diameter and evolution), as no size or temporal information was provided.

Results pt. 7 – Explanation Quality



Prompt: A provider reviews this case. Gender: female. Location: head/neck. Respond with JSON only. Keys: diagnosis, confidence (0-100). Include a runner-up diagnosis.

Model Response:

Model reasoning, summarized for brevity:

- Lesion has asymmetry
- Irregular border and heterogeneous coloring
- High melanoma risk. Biopsy should be performed

Final output:

{"diagnosis": "melanoma", "confidence": 95, "runner_up": "benign nevus"}

Figure 7. The model correctly identifies the image as malignant, and references 3 of 5 ABCDE criteria: asymmetry (A), border (B), and color (C). It also correctly follows the user's instructions to output JSON, which was encouraged by the formatting reward during training.

Training Dynamics

- All charts show a trend of fast early improvement followed by a plateau
- These diminishing returns from continuing training are caused by the fact that the model learns obvious features early on, and has to learn more complex features later in training
- This behavior is common in reinforcement learning

Ablations

Ablating explanation correctness reward

Prompt: What is your diagnosis?
Model Response:
After reviewing the cropped image, I notice that the lesion has irregular borders and asymmetrical shape. The color distribution is heterogeneous, with areas of red and purple. Such characteristics are concerning for melanoma. Although some moles can exhibit similar traits, the combination of these features increases the likelihood of a serious condition. Thus, I recommend urgent intervention.

Figures 9 and 10: Representative responses from models trained without (left) and with (right) explanation supervision (see Appendix A). A lack of explanation supervision allows the model to generate generic descriptions and mix diagnosis with internal reasoning. When explanations are supervised, explanation organization tends to improve, and the model explains why alternative diagnoses can be ruled out.

Prompt: What is your diagnosis?
Model Response:
Diagnosis: malignant
Visual evidence:
- Asymmetry: The lesion is clearly asymmetrical in shape.
- Border irregularity: The edges are uneven and not well-defined.
- Color variation: The pigmentation is uneven, with darker shades concentrated in certain areas.

Reasoning: The combination of significant asymmetry, irregular borders, and heterogeneous pigmentation strongly supports a malignant transformation. The main alternative, a benign nevus, usually exhibits more uniform characteristics and does not show such pronounced abnormalities. Therefore, the evidence leans toward malignancy.

Ablating clipping methods

- Prior work recommends using asymmetric clipping of importance sampling weights, where the upper bound is set higher than the lower bound (Yu et al., 2025)
- When weights were clipped asymmetrically, policy collapse occurred

Figure 8: Red represents asymmetric clipping(between 0.8 and 1.35) and blue represents symmetric clipping (between 0.8 and 1.2). The graph measures accuracy, and towards the end, the asymmetric policy collapses.

Limitations

- Hallucinations sometimes occur relating to details in the image
  - The color in Figure 6 is an example of a hallucination
- The model is designed to be used by clinicians and not by patients, so it is not user-friendly
- The model's accuracy should be refined to improve trust from doctors

Conclusions

- Vision Language Models can be trained with RLVR to accurately classify images and produce detailed explanations
- Improvements from RLVR are statistically significant
- The model is designed as a decision-support tool for doctors, and is not a replacement for medical judgement.

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Appendix

Appendix A: Reward Design

- The formatting reward penalized the model for generating text that could not be parsed by the correctness reward. It also detected certain prompt patterns (i.e. “output JSON”) and penalized the model for violating them
- The correctness reward has two parts:
  - An answer correctness reward, which purely checks if the model’s label matches the ground truth
  - A reward model-based explanation correctness reward, which punishes hallucinations and enforces more nuanced instruction following patterns that could not be enforced by the deterministic formatting reward

Appendix B: Prompt Generation

- During prompt generation, a base template is selected randomly from a handwritten list.
- After this, constraints from different axes, such as output formatting and cautiousness are appended to the prompt at random.
- Each constraint has multiple phrasings, including out of distribution phrasings that are only used during evaluation.
- This prompt generation strategy forces the model to generalize to a wider range of use cases

Appendix C: Curriculum Learning

- In Reinforcement Learning, the model does not gain any new information if it never succeeds in solving a task.
- The data is ordered from the least difficult examples to the most difficult examples to help the model build foundational knowledge before attempting challenging tasks.
- To determine the difficulty of classifying an image, each image in the dataset was given to the untrained model 4 times, and the success rate for the model was logged
- During training, examples were ordered by success rate, with easier samples coming first and harder ones coming towards the end of training.

Appendix D: Infrastructure

- This project was made possible by a $5,000 grant from Thinking Machines Lab to use on their Tinker platform
- The platform made scaling up experiments very straightforward
- The model being trained was Qwen3-VL-30B-A3B-Instruct, and 400 rollouts were performed per step
    - Normally, this would be a headache to implement, but Tinker made it simple

Thank you!